

Privacy Auditing of Synthetic Single-Cell RNA-seq Data

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1 Introduction

Single-cell RNA sequencing (scRNA-seq), named Nature’s Method of the Year in 2013, measures gene expression at cellular resolution, enabling discoveries in transcriptome profiling, tumor microenvironment characterization, and personalized medicine that bulk sequencing cannot [1]. Because scRNA-seq data is linked to individual donors and can reveal sensitive genetic and disease information, it is subject to strict access controls. Even de-identified expression matrices can be re-linked to individuals: one recent study re-identified donors from scRNA-seq counts via eQTL genotypes with 99.9% accuracy [14]. Because genetic variants are heritable, exposure risks harm not only the individual but also biological relatives.

Synthetic data generation (SDG)—training a generative model on real data to produce artificial samples preserving statistical properties—has been proposed for privacy-preserving data sharing [12]. The CAMDA Health Privacy Challenge at ISMB 2025¹ (and again in 2026) formalized this problem on the OneK1K dataset with 1.26 M peripheral blood mononuclear cells from 981 donors [15]. scDesign2 [12], which models per-gene marginal distributions and gene-gene correlations via Gaussian copulas, produced the highest-quality synthetic data under the CAMDA 2025 metrics [8]. A critical question, however, is whether this synthetic data actually protects donor privacy. Membership inference attacks (MIAs) that ask whether a specific individual’s data was used to train a given model are a standard auditing tool in the machine learning privacy literature [11], yet no CAMDA 2025 participant successfully mounted an adversarial MIA against scDesign2. Existing MIAs designed for tabular data [5, 13] fail to perform above random guessing on high-dimensional genomic data [9, 16].

We take a dual perspective on CAMDA 2026 Track II. As **blue-team synthetic scRNA-seq data generators** we submit: (i) **scVI** [7], a VAE with ZINB observation model; (ii) **ZINB-WaVE** [10], a zero-inflated negative binomial factor model; and (iii) **Noisy-scDesign2** ($\epsilon=100$), our novel private adaptation of last year’s Track II/blue team winner. As a **red-team privacy audit** we present **scMAMA-MIA**, the first MIA effective against synthetic scRNA-seq data.

2 Blue Team: Synthetic scRNA-seq Data Generation

Fig. 1 shows UMAP plots of the real OneK1K data (14 cell types) and synthetic samples generated with the SDG methods scDesign2, Noisy-scDesign2, scVI, and ZINB-WaVE, explained below. All synthetic samples mirror the real data distribution well.

Noisy-scDesign2 scDesign2 [12] generates synthetic scRNA-seq counts via a two-group strategy: per-gene, per-cell-type marginals are fit via maximum likelihood over ZINB and Poisson families. Genes with sufficient non-zero expression (“group 2”) are jointly modeled via a Gaussian copula to preserve gene-gene correlations; the remainder (“group 1”) are sampled independently. In preliminary

¹<https://bipress.boku.ac.at/camda2025/the-camda-contest-challenges/>

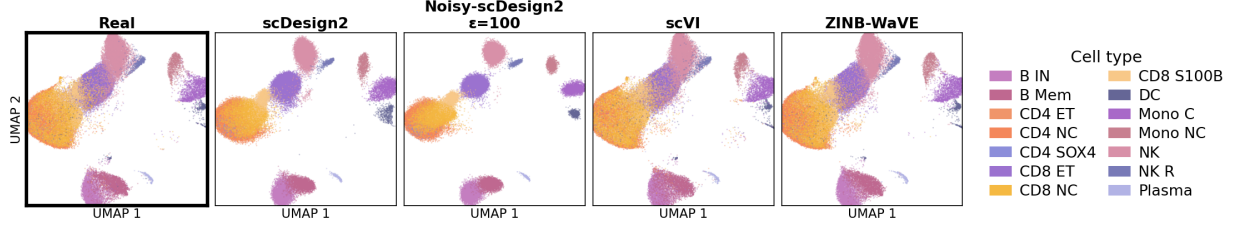


Figure 1: UMAPs for real data samples (50 donors) and the corresponding synthetic data (computed jointly) to illustrate how much information each SDG method is able to preserve.

work [3] we observed that scDesign2 is susceptible to the scMAMA-MIA attack that we present in Section 3. This prompted us to develop Noisy-scDesign2, an adaptation of scDesign2 in which measured gene correlations are perturbed with noise to enhance privacy protection.

Noisy-scDesign2 injects Gaussian noise, calibrated to the donor-level sensitivity of the covariance estimation, into each entry of scDesign2’s per-cell-type covariance matrices. Critically, noise is *not* applied to the one-way marginal parameters, so Noisy-scDesign2 is a **partial, first-line defense** rather than full donor-level differential privacy [2]. Also, the privacy budget ϵ is per cell type. Noised matrices are symmetrized, and projected to the nearest positive semi-definite matrix.

Other SDG methods. **scVI** [7] encodes cells via a variational autoencoder with a ZINB observation model capturing overdispersion and dropout. **ZINB-WaVE** [10] learns a zero-inflated factor model across genes and cells.

3 Red Team: scMAMA-MIA Membership Inference Attack

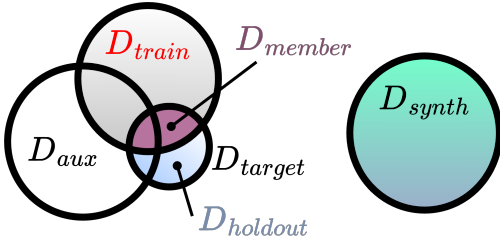


Figure 2: MIA threat model

Assume that a data holder with scRNA-seq dataset D_{train} runs an SDG algorithm to create synthetic scRNA-seq data D_{synth} . In a MIA (see Fig. 2), the adversary has access to D_{synth} but not D_{train} , and must infer for each x in a set of instances D_{target} whether $x \in D_{train}$. As the adversary does not have access to the trained generator, this threat model is considered to be a **black-box (BB)** setting. We use **BB+aux** for the scenario in which an adversary has access to an auxiliary dataset D_{aux} , and **BB-aux** if they don’t.

MAMA-MIA [4] is a class of MIAs that exploits knowledge about statistics-based SDGs to infer membership. To this end, MAMA-MIA identifies which statistics of the training data are preserved (“focal-points”) and aggregates focal-point measurements on target instances to score membership.

scMAMA-MIA. We adapt MAMA-MIA to exploit scDesign2’s Gaussian copula. In the black-box (BB) setting, we run scDesign2 on D_{synth} to obtain a *shadow copula* C_{synth} —a proxy for the copula C_{train} fitted on the hidden training data—per cell type t . The shadow copula provides: (i) a per-cell-type Gaussian copula covariance matrix $\mathcal{R}_t$ over “Group 2” genes; and (ii) marginal distribution parameters for all copula genes (“group 1” and “group 2”). scMAMA-MIA exploits *both* components.

Let F_g^t be the marginal CDF for gene g estimated from the shadow copula C^t , and f_g^t its corresponding PMF. For a target cell x of cell type t :

- **Covariance term (Group 2 genes):** Apply the probability integral transform $u_g = F_g^t(x_g)$, Gaussian-normalize $z_x = \Phi^{-1}(u)$, and compute the Mahalanobis distance:

$$d_s = \sqrt{z_x^\top \mathcal{R}_t^{-1} z_x} \quad (1)$$

The intuition here is that this measures how far all gene–gene pairings for this instance deviate from the expected covariance.

- **Marginal term (Group 1 genes):** Accumulate per-gene log-likelihoods from the shadow copula’s fitted marginals: $\mathcal{B}_x = \sum_{g \in G_1} \log f_g^t(x_g)$.

The combined per-cell membership score is:

$$\lambda_t(x) = \frac{1}{d_s} + \gamma_{\text{eff}} \cdot \mathcal{B}_x, \quad (2)$$

where γ_{eff} is an auto-calibrated weighting. Cell-level scores are averaged per donor and passed through a sigmoid to yield donor-level membership probabilities.

Without auxiliary data, $\lambda = d_s^{-1}$; with D_{aux} the adversary additionally computes d_a by generating a second shadow copula C_{aux} and sets $\lambda = d_a / (d_s + d_a)$. **BB-aux is the most relevant threat model for CAMDA:** adversaries access only the released D_{synth} with no auxiliary real data. (In a *white box* setting, i.e. where the adversary has access to the trained model, the proxy is no longer needed; simply replace C_{synth} with C_{train} .)

4 Experiments

Setup.² We evaluate on **OneK1K** [15] data at the **490-donor** scale (i.e. half of the data), matching the CAMDA 2026 training split, as well as across several smaller size configurations. Results are averaged over 5 random donor splits. We follow [12] and select 1,118 highly variable genes (HVGs) via `sc.pp.highly_variable_genes` in Scanpy with `min_mean=0.0125`, `max_mean=3`, and `min_disp=0.5`.

Fidelity results. We quantify fidelity using LISI (neighborhood mixing between real and synthetic cells; higher is better), ARI (cluster agreement; higher is better), and MMD (distributional distance; lower is better) [6].

At the competition scale of 490 donors, Table 1 shows that scVI and ZINB-WaVE achieve the highest fidelity, followed by scDesign2. Noising scDesign2’s covariance matrix, at $\epsilon = 10$, has no apparent effect on its ARI. However, the noise appears destructive to its LISI and MMD scores, since the donor-level sensitivity grows with cohort size.

On the other hand, when Noisy-scDesign2 is trained on smaller samples of 50 donors³ injecting noise appears not to harm fidelity greatly. Table 2 shows this privacy–utility tradeoff for 50-donor samples, across the full range of ϵ . This preserved fidelity is also illustrated in the UMAPs given in Figure 1, where Noisy-scDesign2 at low epsilon (high noise) has comparable overlap with scDesign2-proper. But in general, both scDesign2 and Noisy-scDesign2’s fidelity isn’t as strong as scVI or ZINB-WaVE’s fidelity, trading fidelity for stronger privacy.

Membership inference attack results. In the right-hand columns of Tables 1 and 2, we provide scMAMA-MIA scores (both $\pm \text{aux}$). For the 490-donor samples, the **-aux** attacks do not outperform random chance. In the **+aux** threat model, the attack does well against the non-private SDG methods, but not against Noisy-scDesign2 (0.59 - 0.68 AUC as opposed to 0.50, chance). These substantial differences in attack accuracies between the two threat models highlight the danger adversaries pose when given access to reference cohorts.

In the smaller, 50-donor setting, at $\epsilon = 10^6$ (weak noise), fidelity is fully preserved and scMAMA-MIA attack AUC is near 0.81; at $\epsilon=10$, the BB+aux attack degrades to 0.63 and the BB–aux attack degrades to 0.52. It likely remains above chance because Group 1 marginal parameters are untouched by the noise. This residual BB–aux signal highlights the limitation of covariance-only DP.

²Code used to conduct the experiments: https://github.com/steveng9/scRNA-seq_privacy_audits.

³50-donor samples have approximately 64k cells, as opposed to approx. 630k cells in the 490-donor samples.

SDG Method	Param	Fidelity			Privacy (scMAMA-MIA AUC)	
		LISI $\uparrow$	ARI $\uparrow$	MMD $\times 10^3 \downarrow$	BB-aux	BB+aux
scDesign2		0.85 $\pm$.00	0.43 $\pm$.03	0.64 $\pm$.42	0.51 $\pm$.02	0.59 $\pm$.03
Noisy-scDesign2	$\epsilon=10$	0.00 $\pm$.00	0.42 $\pm$.02	308.68 $\pm$ 4.51	0.50 $\pm$.03	0.50 $\pm$.03
scVI		0.90 $\pm$.00	0.40 $\pm$.04	0.26 $\pm$.03	0.48 $\pm$.01	0.68 $\pm$.03
ZINB-WaVE		0.89 $\pm$.00	0.40 $\pm$.03	0.32 $\pm$.07	0.50 $\pm$.02	0.61 $\pm$.01

Table 1: **Fidelity and privacy, OneK1K, 490 donors** (mean $\pm$ std, 5 trials).

Privacy Param (ϵ)	LISI $\uparrow$	ARI $\uparrow$	MMD $\times 10^3 \downarrow$	BB-aux	BB+aux
(no noise)	0.87 $\pm$.00	0.48 $\pm$.02	0.45 $\pm$.03	0.63 $\pm$.03	0.79 $\pm$.06
$\epsilon = 10^6$	0.87 $\pm$.00	0.49 $\pm$.01	0.44 $\pm$.02	0.63 $\pm$.03	0.81 $\pm$.07
$\epsilon = 10^5$	0.87 $\pm$.00	0.47 $\pm$.04	0.44 $\pm$.04	0.62 $\pm$.03	0.80 $\pm$.07
$\epsilon = 10^4$	0.85 $\pm$.00	0.46 $\pm$.04	0.42 $\pm$.02	0.59 $\pm$.02	0.79 $\pm$.07
$\epsilon = 10^3$	0.72 $\pm$.01	0.51 $\pm$.03	0.47 $\pm$.02	0.55 $\pm$.03	0.72 $\pm$.05
$\epsilon = 10^2$ (submitted)	0.62 $\pm$.01	0.47 $\pm$.03	0.53 $\pm$.03	0.52 $\pm$.02	0.64 $\pm$.03
$\epsilon = 10$	0.61 $\pm$.01	0.54 $\pm$.02	0.55 $\pm$.01	0.52 $\pm$.02	0.63 $\pm$.02

Table 2: **Privacy-utility tradeoff for Noisy-sDesign2, 50 donors.** (mean $\pm$ std, 5 trials).

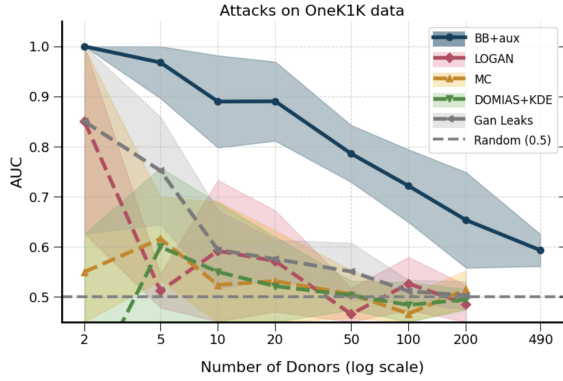


Figure 3: Comparison of scMAMA-MIA with baseline attacks. Bands: ± 1 SD over 5 trials.

Fig. 3 compares donor-level membership inference AUC for scMAMA-MIA (BB+aux) with those of CAMDA 2025 baseline attacks against data generated by scDesign2. Across all training cohort sizes (2, 5, ... 490 donors), scMAMA-MIA substantially outperforms all baseline MIAs. On top of this (with the exception of the LOGAN attack) scMAMA-MIA is $2.6\times$ to $12.3\times$ faster than the others, several of which cannot successfully execute on the 490-donor cohort. Privacy leakage decreases as cohort size grows.

5 Conclusion

We present scMAMA-MIA, the first effective MIA against synthetic scRNA-seq data, and Noisy-scDesign2, a novel differentially private variant of last year’s top CAMDA Track II generator. At 490 donors, scMAMA-MIA reveals meaningful leakage in high-fidelity generators (AUC up to 0.68 for scVI), with disparate impact analysis showing overrepresented demographic subgroups face elevated risk in neural SDG methods. A key interpretive finding: low attack AUC is not a reliable privacy signal when fidelity is also low—lower-quality generators such as NMF appear private only because their outputs are too dissimilar from real data to carry a membership signal. Noisy-scDesign2 partially mitigates leakage but suppresses utility at CAMDA-scale cohort sizes, and extending formal donor-level DP to all scDesign2 parameters (e.g. marginals, covariance and multiple cell types under composition) remains an open problem [2]. As an interim measure, we recommend against releasing copula parameters alongside synthetic data, as white-box access consistently amplifies attack success.

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